

PHOENIX – Threat Analysis Integration Testing Plan

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Project: PHOENIX

Deliverable: Threat Analysis Integration – Integration Testing Plan

1. Purpose

The purpose of this Integration Testing Plan is to define a structured approach for validating the integration between the major components involved in the PHOENIX Threat Analysis Integration.

The testing focuses on verifying that security events can move correctly through the Event Collection Layer, Data Normalisation Layer, Threat Analysis Engine, ADCRS Risk Scoring Engine, Response Decision Manager and Output Layer.

The plan also considers the security controls defined within the Threat Analysis Integration Design Document to ensure that the integration is validated from both functional and cybersecurity perspectives.

2. Testing Objectives

The primary objectives are to verify that:

- Security events are successfully received and validated.
- Event data is correctly normalised into the expected common structure.
- Existing threat-detection rules execute correctly.
- Threat types and severity levels can be identified.
- Event information can be provided to ADCRS for risk scoring.
- ADCRS returns the expected risk score, confidence score and threat classification.
- Threat-analysis and risk-scoring results can be processed by the Response Decision Manager.
- Recommended mitigation and notification requirements can be determined.
- Final threat-analysis results can be generated in the expected structure.

- Dashboard updates can occur successfully.
- Alerts and notifications can be triggered when required.
- Threat-analysis results can be recorded through audit logging.
- Data can be persisted as required by the Output Layer.
- Defined security controls are considered throughout the integration.

3. Components in Scope

The following components are included within the integration testing scope:

1. Input Sources
2. Event Collection Layer
3. Data Normalisation Layer
4. Threat Analysis Engine
5. ADCRS Risk Scoring Engine
6. Response Decision Manager
7. Output Layer

Supported event sources include:

- TEAVS Alerts
- API Events
- User Activity Logs
- IoT Device Events
- System Logs
- Application Events

4. Expected Event Structure

A valid event should follow the expected structure defined by the Threat Analysis Integration design.

```
{  
  "event_id": "EVT-001",  
  "event_type": "login_activity",  
  "timestamp": "2026-08-03T06:30:00Z",  
  "source": "TEAVS",  
  "event_data": {}  
}
```

5. Expected Threat-Analysis Output

The design provides the following example of a standard threat-analysis output:

```
{
  "event_id": "EVT-001",
  "threat_type": "Login Attack",
  "severity": "High",
  "risk_score": 82,
  "confidence_score": 0.91,
  "recommended_action": "lock_account",
  "status": "detected"
}
```

This structure can be used as a reference when validating the integration output.

6. Functional Integration Test Cases

Test ID	Test Scenario	Test Input / Action	Expected Result	Status
FIT-01	Valid Security Event Collection	Submit a valid security event containing the expected event fields	Event is successfully received and passes validation	Not Executed
FIT-02	Event Normalisation	Pass a validated event to the Data Normalisation Layer	Event is converted into the common event structure	Not Executed
FIT-03	Detection Rule Execution	Provide a normalised security event to the Threat Analysis Engine	Relevant existing detection rule executes successfully	Not Executed
FIT-04	Threat Identification	Process an event matching an existing threat-detection rule	Threat type and severity are identified	Not Executed
FIT-05	ADCRS Risk Scoring	Forward the required event information to	ADCRS returns a risk score	Not Executed

		ADCRS		
FIT-06	ADCRS Confidence Scoring	Process event information through ADCRS	Confidence score is returned	Not Executed
FIT-07	Threat Classification	Process event information through ADCRS	Threat classification is returned	Not Executed
FIT-08	Response Decision Processing	Provide threat-analysis and ADCRS results to the Response Decision Manager	Severity, recommended mitigation and notification requirements are determined	Not Executed
FIT-09	Standardised Final Output	Complete the threat-analysis workflow	Final output follows the expected threat-analysis output structure	Not Executed
FIT-10	Dashboard Integration	Complete processing of a threat-analysis result	Threat-analysis result is successfully available for the dashboard update	Not Executed
FIT-11	Alert Generation	Process an event requiring an alert	Alert is generated correctly	Not Executed
FIT-12	Notification Processing	Process an event where notification is required	Required notification is triggered	Not Executed
FIT-13	Audit Logging	Complete threat-analysis processing	Threat-analysis result is recorded for auditing	Not Executed
FIT-14	Data Persistence	Complete threat-analysis processing	Required threat-analysis result is persisted through the Output Layer	Not Executed

7. Security Integration Test Cases

Test ID	Security Control	Test Scenario	Expected Result	Status
SIT-01	JWT Authentication	Attempt to access protected integration functionality without valid authentication	Unauthenticated access is rejected	Not Executed
SIT-02	JWT Authentication	Access protected integration functionality with valid authentication	Authenticated request is permitted subject to authorisation	Not Executed
SIT-03	RBAC	Attempt to access protected functionality using a role without the required permission	Unauthorised role is prevented from accessing the protected functionality	Not Executed
SIT-04	Input Validation	Submit an event that does not meet the expected input requirements	Invalid event is rejected or prevented from continuing through normal processing	Not Executed
SIT-05	Rate Limiting	Generate requests beyond the configured permitted request rate	Excessive requests are controlled according to the implemented rate-limiting behaviour	Not Executed
SIT-06	Secure Error Handling	Trigger an integration error using invalid or failed processing conditions	Error is handled securely without exposing unnecessary sensitive implementation information	Not Executed
SIT-07	Audit Logging	Perform a threat-analysis operation	Relevant threat-analysis activity/result is	Not Executed

			recorded for auditing	
SIT-08	Protected Routes	Attempt unauthorised access to a protected route	Access is prevented	Not Executed
SIT-09	HTTPS Communication	Verify communication for integration interfaces where HTTPS is required	Communication uses HTTPS as required by the integration design	Not Executed

8. Data Validation Test Cases

Test ID	Scenario	Expected Result	Status	Test ID
DVT-01	Valid event_id provided	Event proceeds through validation	Not Executed	DVT-01
DVT-02	Valid event_type provided	Event type is accepted for processing	Not Executed	DVT-02
DVT-03	Valid timestamp provided	Timestamp is accepted within the expected event structure	Not Executed	DVT-03
DVT-04	Valid source provided	Event source is accepted for processing	Not Executed	DVT-04
DVT-05	Event data provided in expected structure	Event can proceed to normalisation/analysis	Not Executed	DVT-05
DVT-06	Required event information is	Event does not continue through	Not Executed	DVT-06

	invalid or unavailable	normal processing without appropriate validation/error handling		
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9. Example End-to-End Integration Test

Test ID

E2E-01

Test Name

Login Attack Threat Analysis Integration

Objective

Verify that a login activity event can progress through the Threat Analysis Integration and produce the expected standardised threat-analysis result.

Example Input

```
{
  "event_id": "EVT-001",
  "event_type": "login_activity",
  "timestamp": "2026-08-03T06:30:00Z",
  "source": "TEAVS",
  "event_data": {}
}
```

Test Flow

Step 1: Submit the security event to the Event Collection Layer.

Step 2: Verify that the event is received and validated.

Step 3: Verify that the event is normalised into the expected common structure.

Step 4: Verify that the Threat Analysis Engine selects and executes the relevant detection logic.

Step 5: Verify that threat type and severity are identified.

Step 6: Verify that the required event information is forwarded to ADCRS.

Step 7: Verify that ADCRS returns risk-scoring information.

Step 8: Verify that the results are processed by the Response Decision Manager.

Step 9: Verify that the expected final threat-analysis result is generated.

Step 10: Verify applicable Output Layer activities such as dashboard update, alert generation, audit logging and data persistence.

Step 11: Verify that a notification is triggered when required.

Expected Example Output

```
{
  "event_id": "EVT-001",
  "threat_type": "Login Attack",
  "severity": "High",
  "risk_score": 82,
  "confidence_score": 0.91,
  "recommended_action": "lock_account",
  "status": "detected"
}
```

Expected Result

The event progresses successfully through the defined Threat Analysis Integration flow and produces the expected standardised threat-analysis output.

Test Status: Not Executed

10. Entry Criteria

Integration testing should begin when:

- Relevant integration components are available for testing.
- Required interfaces between components are available.
- The expected event structure has been confirmed.
- Relevant detection rules are available.
- ADCRS integration required for the selected tests is available.
- The test environment is ready.
- Required security controls relevant to the selected tests have been implemented.

11. Exit Criteria

Integration testing can be considered complete when:

- Required security events can be analysed successfully.
- Relevant detection rules execute correctly.
- ADCRS risk scores are successfully returned.
- Standardised threat-analysis outputs are generated.
- Required dashboard updates occur successfully.
- Alerts and notifications operate as expected.
- Threat-analysis results are recorded for auditing.
- Required security integration tests have been completed.
- Significant integration issues identified during testing have been documented and addressed or formally recorded for further action.

12. Defect Recording

Any issue identified during integration testing should be documented with sufficient information to reproduce and investigate the problem.

Recommended defect information includes:

- Defect ID
- Related Test ID
- Component
- Description
- Steps to Reproduce
- Expected Result
- Actual Result
- Severity
- Evidence
- Status

13. Test Evidence

Evidence should be retained for completed test cases where appropriate.

Evidence may include:

- Test execution screenshots
- Relevant application output
- API request/response evidence where available
- Dashboard results
- Audit-log evidence
- Error-handling results
- Relevant development/testing records

The evidence should clearly identify which test case was executed and whether the actual result matched the expected result.

14. Current Testing Status

This document currently defines the **planned integration testing approach** for the Threat Analysis Integration.

Test cases should remain marked **Not Executed** until the relevant integration has actually been implemented and the corresponding test has been performed.

Test results should not be recorded as Passed or Failed without execution evidence.

15. Summary

This Integration Testing Plan establishes a structured approach for validating the PHOENIX Threat Analysis Integration from both functional and cybersecurity perspectives.

The plan validates the integration flow established in the **Threat Analysis Integration Workflow Diagram** and **Threat Analysis Sequence Diagram**, covering event collection, validation, normalisation, threat detection, ADCRS risk scoring, response decision processing and final output handling.

The plan also incorporates the security controls defined for the integration, including authentication, role-based access control, input validation, rate limiting, secure error handling, audit logging, protected routes and HTTPS communication.

Together, the Integration Workflow Diagram, Threat Analysis Sequence Diagram and Integration Testing Plan provide a consistent set of technical artefacts supporting the future implementation and validation of the PHOENIX Threat Analysis Integration.